

Autosomal Dominant Nonsyndromic Hearing Loss 12 (DFNA12 / DFNA8/12) — Comprehensive Disease Report

Evidence base: synthesized from primary literature (human clinical, mouse model, in vitro, and computational studies). Evidence source type is annotated per claim. PMIDs are given for all key statements.

Summary (Answer to the Research Question)

Autosomal Dominant Nonsyndromic Hearing Loss 12 (DFNA12), now unified with DFNA8 as **DFNA8/12**, is a Mendelian, autosomal-dominant, nonsyndromic sensorineural hearing loss caused by heterozygous, predominantly **missense** mutations in **TECTA** (gene for **α-tectorin**, a major non-collagenous glycoprotein of the cochlear **tectorial membrane**). Mutant α-tectorin is incorporated into and structurally poisons the tectorial membrane (a **dominant-negative** mechanism), degrading the mechanical coupling between the tectorial membrane and the outer-hair-cell stereocilia that drives cochlear amplification and frequency tuning, producing bilateral, usually mid- or high-frequency, prelingual-to-childhood-onset sensorineural hearing loss. The affected α-tectorin **protein domain predicts the phenotype** (ZP domain → mid-frequency, often stable; zonadhesin/ZA domain and cysteine-substituting variants → high-frequency, progressive). DFNA8/12 is one of the most commonly identified single-gene causes of autosomal dominant nonsyndromic hearing loss; management is auditory rehabilitation (hearing aids, and cochlear implantation/electric-acoustic stimulation with favorable outcomes), with no disease-modifying drug.

1. Disease Information

Overview. DFNA12 is a hereditary, nonsyndromic (isolated) sensorineural hearing impairment inherited in an autosomal dominant pattern. The DFNA8 and DFNA12 loci were both mapped to chromosome **11q** and shown to result from mutations in the same gene, **TECTA**; the entity is therefore designated **DFNA8/12** (human clinical/linkage):

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α -Tectorin is "one of the major non-collagenous components of the tectorial membrane" of the inner ear

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Key identifiers. - **OMIM (phenotype):** 601543 (Deafness, autosomal dominant 12; encompassing DFNA8/12) - **OMIM (gene TECTA):** 602574 - **Gene:** TECTA — HGNC:11720; NCBI Gene 7007; Ensembl ENSG00000109927; UniProt **O75443** (α -tectorin, 2155 aa) - **MONDO:** DFNA12 corresponds to MONDO "autosomal dominant nonsyndromic hearing loss 12" (also captured under the DFNA8/12 concept); TECTA-related dominant deafness - **Orphanet:** included within "Rare genetic deafness"/autosomal dominant nonsyndromic sensorineural deafness type DFNA - **ICD-10:** H90.5 (sensorineural hearing loss, unspecified) / H90.3 (bilateral). **ICD-11:** AB52 (sensorineural hearing loss) - **MeSH:** related terms "Hearing Loss, Sensorineural"; "Deafness"; gene "TECTA / tectorin alpha"

Synonyms / alternative names: DFNA8; DFNA12; DFNA8/12; TECTA-related autosomal dominant nonsyndromic hearing loss; deafness, autosomal dominant 8/12; α -tectorin-related dominant deafness.

Information source: aggregated disease-level resources (OMIM, published pedigrees/cohorts) and individual multigenerational family studies; not derived from population EHR.

2. Etiology

Causal factors — genetic. DFNA12 is a **monogenic** disorder. The sole established cause is a heterozygous pathogenic variant in **TECTA**. The initiating lesion is almost always a **missense** substitution (occasionally an in-frame splice variant) affecting conserved residues of α -tectorin

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"In both families, mutation analysis revealed missense mutations which replace conserved amino-acid residues within the zona pellucida domain of TECTA"

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Genetic risk factors. The causal variant itself is the risk factor; there are no separate susceptibility loci for the Mendelian form. The specific **domain** affected is the principal modifier of expression (see §4). No environmental modifiers are established.

Environmental risk factors. None established as causing DFNA12. As with any sensorineural hearing loss, generic aggravators (noise exposure, ototoxic drugs, aging) could additively worsen hearing; notably, in the Tecta^{C1509G/+} mouse, noise exposure caused incomplete recovery and increased outer-hair-cell loss versus wild type (model organism:

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suggesting **gene–environment interaction** whereby a defective tectorial membrane increases vulnerability to noise. TECTA is also among genes implicated in age-related (multifactorial) hearing loss

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Protective factors. None specifically identified. A single functional TECTA allele is sufficient for near-normal hearing (DFNB21 heterozygous carriers are unaffected), which is why complete loss-of-function is only pathogenic when biallelic

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Gene–environment interaction. Inferred increased susceptibility to noise-induced damage on a mutant-TECTA background (model organism evidence:

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direct human data limited.

Infectious agents: not applicable.

3. Phenotypes

Core phenotype: bilateral, symmetric, nonsyndromic **sensorineural hearing loss** with no vestibular, visual, renal, or other systemic involvement.

Phenotype	Type	HPO term	Characteristics / frequency
Sensorineural hearing impairment	clinical sign	HP:0000407	Defining feature, ~100% of affected
Bilateral sensorineural hearing impairment	clinical sign	HP:0008619	Bilateral, usually symmetric
Mid-frequency sensorineural hearing impairment ("cookie-bite"/U-shaped audiogram)	clinical sign	HP:0410007 / HP:0008542	Typical of ZP-domain variants (P 21520338 P 37927186 P 9763681)
High-frequency hearing impairment	clinical sign	HP:0000399 / HP:0008565	Typical of ZA-domain variants (P 21520338 P 24363064)
Congenital / prelingual sensorineural hearing impairment	clinical sign	HP:0008527	Many families prelingual/ congenital (P 9763681)
Progressive sensorineural hearing impairment	clinical sign	HP:0000408	With cysteine-substituting / ZA variants (P 21520338 P 24363064)

Age of onset: congenital/prelingual to childhood; some late-onset/progressive forms present in adulthood

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Severity: mild to severe (moderate-to-severe common); "moderate to severe... U-shaped form with maximum loss at 2,000 Hz" in the original DFNA8 family
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Progression: frequently **stable/nonprogressive** for ZP non-cysteine variants; **progressive** for cysteine-substituting and ZA-domain variants

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Frequency among affected: hearing loss is fully penetrant in most reported families; configuration/severity vary by genotype.

Quality-of-life impact: hearing loss impairs speech perception, communication, and (for prelingual cases) spoken-language acquisition and education; early amplification mitigates these effects. No disease-specific QoL instrument data (EQ-5D/SF-36) are available specifically for DFNA12; general SNHL QoL literature applies.

4. Genetic / Molecular Information

Causal gene: **TECTA** (α -tectorin), 11q23.3; OMIM 602574; HGNC:11720; NCBI Gene 7007; UniProt O75443. Encodes a large secreted, GPI-anchored (during trafficking) modular glycoprotein of 2155 aa with an **entactin/nidogen-G1-like (NIDO) domain**, three **von Willebrand factor type D (vWFD1–3)** repeats within a **zonadhesin-like (ZA)** region, and a C-terminal **zona pellucida (ZP)** domain

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Pathogenic variants. - **Variant type/class:** predominantly **missense** in DFNA8/12; occasionally in-frame/aberrant **splice** variants (e.g., c.5999G>A p.Gly2000Glu causing exon-20 mis-splicing; c.5383+6T>A causing exon-16 skipping)

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Truncating variants (nonsense/frameshift/splice/large deletions) instead cause **recessive DFNB21**

([P 17431902](#)

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 - **Distribution across domains:** mutations occur in **all domains** — entactin/NIDO, vWFD1–3, D1–D2 and TIL2 connectors, and ZP
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 - **Representative variants:** p.Cys1509Gly (C1509G), p.Cys1619Ser (ZA), p.Leu1820Phe+Gly1824Asp and p.Cys1837Gly (ZP), p.Thr1866Met, p.Arg1890Cys, p.Cys1036Tyr, p.Val317Glu, near-ZP c.6183G>T
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 - **Classification (ACMG/AMP):** pathogenic/likely pathogenic when segregating and absent from controls; segregation analysis is often decisive for reclassifying VUS (e.g., c.6183G>T upgraded to likely pathogenic;
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 - **Allele frequency:** pathogenic DFNA8/12 missense alleles are rare/absent in gnomAD and matched controls
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 - **Origin: germline;** de novo cases possible but most are familial. Not somatic. - **Functional consequence: dominant-negative** for DFNA8/12 missense alleles (mutant α -tectorin incorporated into TM disrupts its assembly); **loss-of-function** for recessive DFNB21 alleles
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Modifier genes: the affected α -tectorin domain is the dominant determinant of expressivity
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 Interacting TM proteins (β -tectorin/**TCTB**, **CEACAM16**, **OTOG/OTOGL**, collagen II) could theoretically modify phenotype; **CEACAM16** co-immunoprecipitates with α -tectorin and its

mutation causes DFNA4
([P 21368133](#)).

No formal human modifier-gene study for DFNA12.

Epigenetic information: none established for DFNA12.

Chromosomal abnormalities: point mutations, not aneuploidy/translocations. Large intragenic deletions (e.g., exon-10 deletion) occur in the recessive form and require CNV-aware analysis

([P 17431902](#)).

Historically, TECTA haploinsufficiency was proposed to contribute to hearing loss in some Jacobsen-syndrome (11q deletion) cases

([P 9503015](#)).

5. Environmental Information

DFNA12 is a purely genetic disorder; no environmental, lifestyle, or infectious agent causes it. Generic exacerbating exposures (loud noise, ototoxic aminoglycosides/cisplatin, aging) may additively worsen hearing, and mouse data indicate a defective tectorial membrane increases susceptibility to noise-induced outer-hair-cell loss (model organism:

([P 21567249](#)).

No dietary/occupational protective or causal factors are established.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **TECTA missense (or in-frame splice) mutation** alters a conserved residue in a specific α -tectorin domain (entactin/NIDO, vWFD/ZA, or ZP) — **results in** a structurally abnormal α -tectorin monomer (human genetics:

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splicing:

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2. The mutant α -tectorin is secreted by cochlear supporting/interdental cells and **incorporated into the assembling tectorial membrane**, where it **acts dominant-negatively** on α -tectorin self-assembly and its interactions with β -tectorin and type II collagen (inferred from unaffected DFNB21 heterozygotes vs affected DFNA8/12 heterozygotes:

P 9949200

TM assembly:

P 26806019

3. This **leads to a domain-specific structural defect** of the tectorial membrane — e.g., ZP-domain mutations vs ZA-domain mutations "generate distinctly different changes in the structure of the TM" (loss/disorganization of the striated-sheet matrix, altered collagen crosslinking, shortening, or detachment) (model organism:

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4. **Branch A (ZP domain):** predominantly disrupts the mid/apical TM → **mid-frequency** loss, often **stable**

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5. **Branch B (ZA/zonadhesin domain or cysteine substitution):** disrupts basal TM/covalent crosslinking → **high-frequency** and/or **progressive** loss

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6. The abnormal TM **results in** defective mechanical coupling to the **outer-hair-cell (OHC) stereocilia** — e.g., a shortened TM contacts only the first OHC row and **increases shear force on those stereocilia by ~50%** (computational + model organism:

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7. This **impairs the cochlear amplifier**: the TM normally "ensures that outer hair cells can effectively respond to basilar membrane motion and that feedback is delivered with the appropriate gain and timing required for amplification"; when α -tectorin is defective the cochlea is **~35 dB less sensitive** (model organism:

P 11087000

8. Loss of active amplification and frequency selectivity **results in elevated auditory thresholds** in the affected frequency band — clinically, **bilateral sensorineural hearing loss** (human:

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9. **Downstream/secondary (inferred)**: increased OHC vulnerability to noise and altered prestin expression may add a slowly progressive component in some genotypes (model organism:

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Upstream vs downstream: upstream = mutant protein + TM matrix defect (primary lesion in an acellular extracellular matrix); downstream = OHC mechanotransduction/amplification failure and threshold elevation. The **spiral ganglion/auditory nerve and hair-cell bodies are largely preserved**, which is therapeutically important (basis for good cochlear-implant outcomes).

Molecular pathways / processes. This is primarily an **extracellular-matrix (ECM) assembly** disorder rather than a classical signaling-cascade disease. Relevant GO biological processes: **sensory perception of sound (GO:0007605)**, **detection of mechanical stimulus involved in sensory perception of sound (GO:0050910)**, **inner ear morphogenesis (GO:0042472)**, **extracellular matrix organization (GO:0030198)**, **tectorial membrane development**. Cellular component: **extracellular matrix (GO:0031012)** / tectorial membrane. No apoptosis/inflammation/immune or metabolic pathway is centrally implicated; late OHC loss (model) would proceed via mechanical stress rather than a defined death pathway.

Protein dysfunction. Structural (dominant-negative) rather than enzymatic; α -tectorin has no catalytic activity — it is a structural ECM glycoprotein. Cysteine substitutions disrupt disulfide-mediated crosslinking, correlating with progressive phenotypes (P 21520338).

Immune involvement / metabolic / epigenetic: not implicated.

Cell types (CL): outer hair cell (CL:0000601), inner hair cell (CL:0000589), cochlear supporting/interdental cells (tectorin-secreting).

Molecular profiling: no human transcriptomic/proteomic/metabolomic disease signatures reported; mechanistic data derive from mouse cochlear physiology, immunogold/freeze-etch ultrastructure

(P 26806019),
and atomic-force microscopy of TM mechanics
(P 25564867).

7. Anatomical Structures Affected

- **Organ level:** the **inner ear (UBERON:0001846)**, specifically the **cochlea (UBERON:0001844)** — the auditory (nervous/special-sense) system. No secondary organ involvement; **vestibular function is spared** (nonsyndromic).
- **Primary structure:** the **tectorial membrane (UBERON:0002233)**, an acellular ECM within the **organ of Corti (UBERON:0002227)**, and its attachment to the **spiral limbus** (P 26806019).
- **Tissue/cell level:** sensory epithelium of the organ of Corti; **outer hair cells (CL:0000601)** and **inner hair cells (CL:0000589)** are functionally affected via the abnormal overlying TM; tectorin-secreting **supporting/interdental cells** produce the defective matrix.
- **Subcellular (GO-CC): extracellular matrix (GO:0031012)** / tectorial membrane; α -tectorin is a secreted glycoprotein.
- **Localization / lateralization:** **bilateral**, usually symmetric (HP:0008619).

8. Temporal Development

- **Onset:** typically **congenital/prelingual to early childhood**; described as "congenital, nonprogressive" in the founder DFNA8 family

(P 9763681).

Some genotypes present later (adult/late-onset), overlapping age-related hearing loss

(P 42379497).

- **Onset pattern:** insidious/chronic (not acute).
- **Progression:** **stable/nonprogressive** for many ZP non-cysteine variants; **slowly progressive** for cysteine-substituting and ZA-domain variants

(P 21520338).

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Progression rate is slow when present.

- **Disease course:** chronic, lifelong; not episodic/relapsing.
- **Remission:** none spontaneously; only functional improvement with amplification/implantation.
- **Critical period:** the window for spoken-language acquisition in prelingual cases makes **early identification (newborn screening) and early amplification** essential.

9. Inheritance and Population

- **Inheritance:** **autosomal dominant** (MONDO/OMIM 601543); recessive **DFNB21** (OMIM 603629) at the same locus from biallelic LOF alleles

(P 9949200).

- **Penetrance:** generally **high/complete** in reported families (clear multigenerational segregation; e.g., 5-generation pedigree,

(P 40583560).

- **Expressivity: variable and domain-dependent** (frequency band, severity, progression)

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- **Genetic anticipation:** none (not a repeat-expansion disorder).
- **Germline mosaicism:** not specifically documented; de novo variants possible.
- **Founder effects:** confirmed for recurrent alleles — "four of them—p.Cys1036Tyr, p.Cys1837Gly, p.Thr1866Met, and p.Arg1890Cys—were observed in more than one unrelated family. For two of these mutations founder effects were also confirmed"

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- **Consanguinity:** relevant to recessive DFNB21, not to dominant DFNA12.
- **Carrier frequency:** pathogenic dominant alleles are rare; not a routine carrier-screening target.
- **Epidemiology:** no precise prevalence figure exists for DFNA12 specifically. Background: hearing impairment affects **~1 in 500 newborns**, and ~80% of genetic hearing loss is nonsyndromic

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DFNA8/12 is "**the most identified subtype of nonsyndromic autosomal dominant hearing loss**"

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and a leading cause among molecularly solved **adult-onset** dominant SNHL (top gene, 4/15 solved cases,

P 36190904
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- **Population/geography:** reported worldwide (Spanish, Belgian, English, Austrian, French, Korean, Chinese, Iranian, Croatian, etc.); **no strong ethnic predilection**; specific founder alleles are population-enriched.
- **Sex ratio:** ~1:1 (autosomal). **Age distribution:** all ages (onset childhood; diagnosis across the lifespan).

10. Diagnostics

- **Audiologic testing (clinical signs):** pure-tone audiometry (mid-frequency "cookie-bite"/ U-shaped or high-frequency SNHL), tympanometry (normal middle ear), otoacoustic emissions and auditory brainstem response (ABR) — abnormal, consistent with cochlear/ OHC dysfunction; newborn OAE/ABR screening detects congenital cases.
- **Laboratory/biomarkers: none;** no blood/urine biomarker. Diagnosis is audiologic + genetic.
- **Imaging:** temporal-bone CT/MRI is typically **normal** (no malformation) — used to exclude structural/other causes.
- **Genetic testing (definitive):** recommended approach is **NGS**. Comprehensive **hearing-loss gene panels** or **whole-exome/whole-genome sequencing** including TECTA; single-gene TECTA testing when the audiogram (mid-frequency, dominant family history) is suggestive. **CNV/deletion analysis** needed to detect large intragenic deletions. Diagnostic yield of NGS panels/WES in bilateral SNHL is ~**20–48%**

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"48% (50/105) of patients were genetically diagnosed";

P 36804529
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39.5%; adult SNHL 23% with TECTA the top gene,

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Karyotype/CMA/FISH/mtDNA/repeat-expansion testing are **not** indicated for isolated DFNA12.

- **Clinical criteria:** diagnosis = bilateral nonsyndromic SNHL + AD family history + a pathogenic/likely-pathogenic heterozygous TECTA variant with segregation (ACMG/AMP).
- **Differential diagnosis:** other mid-frequency dominant SNHL genes (**EYA4/DFNA10**, **COL11A2/DFNA13**, **CEACAM16/DFNA4**, **WFS1/DFNA6/14/38 [low-frequency]**, **DIAPH1**, **P2RX2/DFNA41**, **KCNQ4/DFNA2**)

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exclude GJB2/GJB6, syndromic causes, noise/ototoxic/age-related loss.

- **Screening/cascade:** newborn hearing screening; **cascade genetic testing** of at-risk relatives once the familial variant is known.

11. Outcome / Prognosis

- **Survival/mortality:** DFNA12 is **not life-limiting**; normal life expectancy; no disease-specific mortality.
- **Morbidity/disability:** the burden is **communication disability** from bilateral SNHL; magnitude depends on severity/onset. For prelingual cases, untreated loss impairs language and educational attainment; early intervention markedly improves outcomes.
- **Disease course:** chronic and lifelong; often stable (ZP) or slowly progressive (ZA/cysteine variants).
- **Recovery:** no spontaneous recovery of hearing; functional recovery is achievable with hearing aids or cochlear implantation/EAS.
- **Prognostic factors:** the **affected α -tectorin domain / variant type** is the key prognostic marker for audiometric configuration and progression

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 preserved neural elements predict good CI outcomes
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 Quality-of-life outcomes track with device benefit; no molecular prognostic biomarker beyond genotype.

12. Treatment

No **pharmacological, gene, cell, RNA, or curative therapy** exists for DFNA8/12. Management is **auditory (re)habilitation**, genotype-informed.

- **Amplification (first-line, mild–severe): hearing aids** (NCIT: Hearing Aid) and assistive listening devices.

- **Cochlear implantation (severe–profound):** effective; because the primary lesion is in the acellular TM with **preserved spiral ganglion/auditory nerve**, implantation restores useful hearing. TECTA patients "showed relatively good auditory performance with CI including EAS"
([P 24130743](#)).
- **Electric-acoustic stimulation (EAS):** appropriate where useful **residual low-frequency hearing** remains (common with mid/high-frequency-predominant TECTA loss)
([P 24130743](#)).
- **Rehabilitation/supportive: speech-language therapy**, auditory training, educational support, sign-language/communication options as chosen (NCIT: Speech Therapy; Auditory Rehabilitation).
- **Pharmacogenomics:** not applicable.
- **Experimental/future:** no DFNA12-specific trials; general inner-ear **gene therapy / antisense (allele-specific knockdown of a dominant-negative allele)** are conceptual future directions but **not yet clinical** for TECTA.
- **Treatment strategy:** severity- and genotype-guided algorithm — early amplification → CI/ EAS if amplification insufficient → lifelong audiologic follow-up. **CHEBI drug entities: none applicable.**

13. Prevention

- **Primary prevention:** the disorder cannot be prevented in mutation carriers; **genetic counseling** (50% offspring risk per affected heterozygote) and reproductive options—**preimplantation genetic testing (PGT-M)** or **prenatal diagnosis** when the familial variant is known—can prevent transmission.
- **Secondary prevention: universal newborn hearing screening** (OAE/ABR) plus **cascade genetic testing** enables early diagnosis and early amplification during the critical language-acquisition window.
- **Tertiary prevention:** early/optimal amplification or implantation to prevent secondary language, educational, and psychosocial complications; **avoid additive insults** (noise protection, avoid ototoxic drugs), supported by mouse evidence of heightened noise

vulnerability

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- **Counseling:** genetic counseling per NSGC/ACMG; **immunization/public-health/environmental** measures are not applicable to this genetic condition.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** **Mouse** *Mus musculus* (NCBI Taxon 10090), gene **Tecta** (NCBI Gene 21683), on chromosome 9 (syntenic to human 11q;

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Orthologs exist in rat, zebrafish, and other vertebrates; α -tectorin and the tectorial membrane are evolutionarily conserved across mammals.

- **Natural disease:** no well-characterized spontaneous companion-animal or wildlife TECTA deafness is prominently reported in OMIA for DFNA12; the disease is studied via engineered mouse models rather than natural animal disease.
- **Comparative biology:** the mouse faithfully models human TM biology; the **genotype–phenotype correlation is conserved** (ZP vs ZA domain → distinct TM defects and audiometric configurations;

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A species-specific incidental finding is **audiogenic seizure susceptibility** in Tecta knock-in mice, not seen in humans

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- **Transmission:** not applicable (non-infectious, non-zoonotic).

15. Model Organisms

Primary model: mouse (mammalian). Faithful recapitulation of human DFNA8/12 mechanism and genotype–phenotype correlation.

- **Tecta knockout (targeted deletion):** TMs **detached** from the cochlear epithelium, lacking all non-collagenous matrix; basilar membrane still tuned but **35 dB less sensitive**; abnormal cochlear microphonic phase/symmetry — established the TM's role in cochlear-amplifier gain/timing

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Models the recessive/LOF (DFNB21) situation and TM function.

- **Knock-in point-mutation lines (model dominant DFNA8/12):** **Tecta^{L1820F,G1824D}/+** and **Tecta^{C1837G}/+** (ZP-domain; stable and progressive mid-frequency loss, respectively) and **Tecta^{C1619S}/+** (ZA-domain; progressive high-frequency loss). "Mutations in the ZP and ZA domains generate distinctly different changes in the structure of the TM"; ABR thresholds elevated 30–40 dB (ZP) and 20–30 dB (ZA) over 8–40 kHz

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- **Tecta^{C1509G}/+:** shortened TM contacting only the first OHC row, ~50% increased stereocilia shear force, increased noise vulnerability/OHC loss

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- **Tecta/Tectb double mutant:** fully **detached** TM; used to show ECM regulation of OHC Ca²⁺, MET currents, and otoacoustic emissions

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Model characteristics. Strengths: reproduce domain-specific TM ultrastructural defects, threshold elevations, and stability/progression matching human genotypes; enable cochlear micromechanics, OAE/ABR, and AFM studies. **Limitations:** murine hearing frequency range differs from human; incidental audiogenic seizures; long-term human progression not fully captured. **Resources:** MGI (Tecta), IMPC/IMSR for strain availability.

Supported and Refuted Hypotheses

Supported: - H1: DFNA12 is caused by heterozygous (mostly missense) TECTA mutations — **supported**

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- H2: The affected α -tectorin domain predicts audiometric phenotype/progression — **supported** in humans and mice

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- H3: DFNA8/12 missense alleles act by a dominant-negative mechanism (vs LOF recessive DFNB21) — **supported**

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- H4: The tectorial-membrane defect impairs cochlear amplification, elevating thresholds — **supported**

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- H5: Neural elements are preserved, enabling good CI/EAS outcomes — **supported**

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Refuted / not supported: - That TECTA haploinsufficiency alone causes dominant deafness — **refuted** for the classic form (heterozygous LOF carriers are unaffected; dominance requires a dominant-negative missense allele)

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- That DFNA12 involves vestibular, systemic, immune, or metabolic dysfunction — **not supported** (strictly nonsyndromic/cochlear).

Limitations and Future Directions

- **Epidemiology:** no precise prevalence/incidence for DFNA12 specifically; figures are inferred from ADNSHL cohorts.
 - **Omics:** no human transcriptomic/proteomic/metabolomic disease signatures; mechanism rests on mouse physiology and ultrastructure.
 - **Penetrance/modifiers:** quantitative penetrance and formal modifier-gene studies are lacking.
 - **Therapeutics:** no molecular therapy; **allele-specific silencing or gene-editing of dominant-negative alleles** and inner-ear gene therapy are logical but unproven future directions.
 - **Human structural biology:** no experimental α -tectorin structure; AlphaFold models (UniProt O75443) could refine variant-effect prediction.
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Key References (PMID)

9590290 (gene discovery); 9763681, 9503015 (locus mapping); 21520338 (mutation spectrum & genotype–phenotype/founder effects); 24363064 (three knock-in mouse models); 11087000 (Tecta KO – cochlear amplification); 21567249 (C1509G shortened TM mechanics); 26806019 (TM assembly/crosslinking); 33559882 (Tecta/Tectb detached TM); 25564867 (TM striated-sheet mechanics); 9949200 (dominant-negative vs DFNB21 LOF); 17431902, 18022253, 27368438, 28012541 (DFNB21 LOF alleles); 20947814, 37927186, 40583560 (human variants/segregation/splicing); 21368133 (CEACAM16– α -tectorin/DFNA4); 36190904, 38224868, 36804529, 24130743 (diagnostics & CI/EAS); 25281338, 42379497 (epidemiology/ARHL context); 25809937 (mid-frequency SNHL differential).
